Python Classes and Objects – Class work

1. Student Information System (Basic)

Problem Statement:

Create a Student class to store the student's name, roll number, and marks obtained in three subjects.

Implement methods to:

- Accept student details.
- Calculate the total marks.
- Calculate the percentage.
- Display the complete student report.

Sample Output:

Name : Ananya
Roll No : 101
Total Marks: 255
Percentage : 85.0%

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2. Rectangle Calculator (Basic)

Problem Statement:

Create a Rectangle class with attributes length and breadth.

Implement methods to:

- Calculate the area.
- Calculate the perimeter.
- Display the dimensions and results.

Sample Output:

Length : 12
Breadth : 8
Area : 96
Perimeter : 40

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3. Bank Account Management (Basic–Intermediate)

Problem Statement:

Design a BankAccount class to manage customer accounts.

Implement methods to:

- Deposit money.
- Withdraw money.
- Check balance.
- Display account details.

Ensure that withdrawal is not allowed if the amount exceeds the available balance.

Sample Output:

Deposited Successfully.
Current Balance: ₹25000

Withdrawal Successful.
Remaining Balance: ₹20000

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4. Employee Salary Management (Intermediate)

Problem Statement:

Create an Employee class containing employee ID, name, and monthly salary.

Implement methods to:

- Display employee details.
- Calculate annual salary.
- Increase salary by a given percentage.

Sample Output:

Employee Name : Rohan
Monthly Salary: ₹50000
Annual Salary : ₹600000
Updated Salary: ₹55000

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5. Book Library System (Intermediate)

Problem Statement:

Create a Book class with attributes:

- Book ID
- Title
- Author
- Availability Status

Implement methods to:

- Issue a book.
- Return a book.
- Display book details.

Prevent issuing a book that is already issued.

Sample Output:

Book Issued Successfully.
Availability Status: Not Available

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6. Shopping Cart Application (Intermediate)

Problem Statement:

Create a Product class containing product name, quantity, and price per unit.

Implement methods to:

- Calculate total price.
- Update quantity.
- Display product details.

Sample Output:

Product Name : Laptop
Quantity : 2
Unit Price : ₹45000
Total Price : ₹90000

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7. Vehicle Rental System (Intermediate)

Problem Statement:

Design a Vehicle class containing:

- Vehicle Number
- Vehicle Type
- Rent per Day

Implement methods to:

- Accept vehicle details.
- Calculate total rental amount based on the number of days rented.
- Display the bill.

Sample Output:

Vehicle Type : Car
Days Rented : 5
Total Rent : ₹10000

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8. Mobile Phone Inventory (Intermediate)

Problem Statement:

Create a MobilePhone class to store:

- Brand Name
- Model Name
- Price
- Available Stock

Implement methods to:

- Display phone details.
- Sell a specified quantity of phones.
- Update stock after sale.

Display an appropriate message if sufficient stock is unavailable.

Sample Output:

Sale Successful.
Remaining Stock: 12

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9. Electricity Bill Generator (Intermediate)

Problem Statement:

Create an ElectricityBill class containing:

- Consumer Name
- Consumer Number
- Units Consumed

Implement methods to:

- Calculate electricity charges using the following slab:

Units	**Rate**
First 100 units	₹5/unit
Next 100 units	₹7/unit
Above 200 units	₹10/unit

- Display the final bill.

Sample Output:

Consumer Name : Amit
Units Consumed: 250
Total Bill : ₹1700

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10. Movie Ticket Booking System (Intermediate)

Problem Statement:

Design a MovieTicket class containing:

- Movie Name
- Ticket Price
- Number of Seats Available

Implement methods to:

- Book tickets.
- Cancel tickets.

- Display booking details.
- Update seat availability.

Do not allow booking if requested seats exceed availability.

Sample Output:

Booking Successful.
Tickets Booked : 4
Amount Payable : ₹800
Seats Remaining: 46